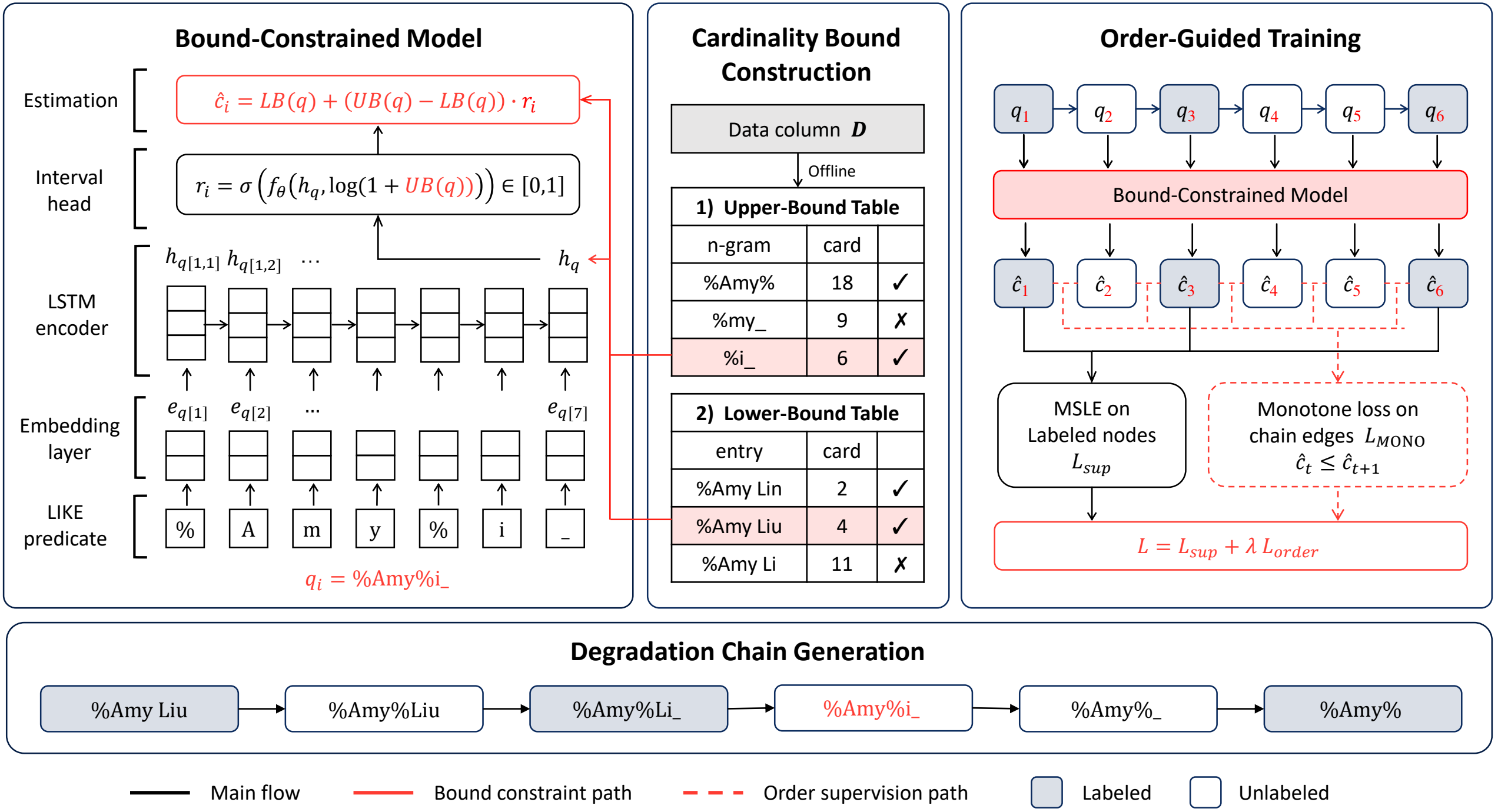

Bound-Constrained Model
Estimation
$\hat{c}_i = LB(q) + (UB(q) - LB(q)) \cdot r_i$
Interval head
$r_i = \sigma\left(f_\theta\left(h_q, \log(1 + UB(q))\right)\right) \in [0,1]$
LSTM encoder
$h_{q[1,1]}$ $h_{q[1,2]}$ ⋯ h_q
Embedding layer
$e_{q[1]}$ $e_{q[2]}$ ⋯ $e_{q[7]}$
LIKE predicate
% A m y % i _
q_i = %Amy%i_
Cardinality Bound Construction
Data column D
Offline
1) Upper-Bound Table
n-gram | card |
%Amy% | 18 | ✓
%my_ | 9 | ✗
%i_ | 6 | ✓
2) Lower-Bound Table
entry | card |
%Amy Lin | 2 | ✓
%Amy Liu | 4 | ✓
%Amy Li | 11 | ✗
Order-Guided Training
q_1 q_2 q_3 q_4 q_5 q_6
Bound-Constrained Model
$\hat{c}_1$ $\hat{c}_2$ $\hat{c}_3$ $\hat{c}_4$ $\hat{c}_5$ $\hat{c}_6$
MSLE on Labeled nodes L_{sup}
Monotone loss on chain edges L_{MONO} $\hat{c}_t \le \hat{c}_{t+1}$
$L = L_{sup} + \lambda L_{order}$
Degradation Chain Generation
%Amy Liu
%Amy%Liu
%Amy%Li_
%Amy%i_
%Amy%_
%Amy%
Main flow
Bound constraint path
Order supervision path
Labeled
Unlabeled